STANDARD OPERATING PROCEDURE (SOP)

VLAN and Switch Port Configuration SOP

SOP ID	NET-VLAN-SW-001	**Version**	1.0
Status	Approved Draft	**Environment**	SOC.lab
Department	Network Engineering / Security Engineering	**Owner**	Infrastructure Engineering

Purpose. This SOP establishes the standard design, implementation, validation, and maintenance process for VLAN segmentation, subnet allocation, and switch port configuration within SOC.lab. It combines logical network segmentation and switch interface standards into a single controlled procedure so that new networks, access ports, and trunk uplinks are deployed consistently, securely, and with repeatable change control.

1. Scope

This SOP applies to Layer 2 switching and VLAN-based segmentation supporting user, server, security-monitoring, and management networks in SOC.lab. It covers VLAN creation, subnet assignment, switch port role selection, access port configuration, trunk configuration, labeling, validation, troubleshooting, rollback, and documentation.

2. Roles and Responsibilities

Role	Primary Responsibility	Typical Tasks
Network Engineer	Design and deploy VLAN/subnet structure	Create VLANs, assign subnets, configure trunks and access ports, validate routing dependencies
Security Engineer	Review segmentation and least-privilege network boundaries	Confirm zone separation, review management access restrictions, verify logging and control coverage
Systems / Operations	Coordinate endpoint and server cutover	Update gateway/IP settings, test service connectivity, confirm change windows and dependency impacts
Change Approver	Authorize production network changes	Review implementation plan, rollback plan, validation evidence, and risk impact

No unmanaged ad hoc VLAN or switch port change is permitted without a documented purpose, target subnet, owner, and validation record.

3. Design Standards

3.1 Segmentation Principles

- Use VLANs to separate management, user, security-monitoring, server, and lab traffic rather than permitting a flat broadcast domain.
- Assign one defined business or security purpose per VLAN.
- Pair each production VLAN with a documented IPv4 subnet, default gateway, DHCP model, and owner.

- Avoid using temporary or undocumented trunk allowances in production.
- Apply least privilege to switch management access and trunk VLAN allowances.

3.2 Standard Network Artifacts

Artifact	Minimum Required Field(s)	Owner	Update Trigger
VLAN register	VLAN ID, name, purpose, subnet, gateway, DHCP model, owner	Network Engineering	New VLAN, decommission, subnet change
Port map	Switch, interface, mode, assigned VLAN(s), endpoint/system, description	Network Engineering	Move/add/change of endpoint or uplink
Subnet inventory	Subnet, mask, gateway, DNS/DHCP notes, reserved ranges	Infrastructure Engineering	Any address plan change
Change record	Reason, approver, rollback steps, validation results	Change Owner	Every production implementation

3.3 Naming and Numbering Standard

VLAN names should be concise, role-based, and stable. Example pattern: SOC-MGMT, SOC-LAN, SOC-SIEM, SOC-SERVER, SOC-LAB, SOC-USER. VLAN IDs and subnets must be reserved in advance and not reused without formal retirement documentation.

4. VLAN and Subnet Implementation Procedure

4.1 Pre-Implementation Requirements

1. Confirm the business and security reason for the new or changed VLAN.
2. Assign the VLAN ID, network name, subnet, default gateway, and DHCP/static model.
3. Identify inter-VLAN routing requirements and firewall dependencies.
4. Identify the switches, uplinks, access ports, and endpoints affected.
5. Prepare a rollback plan before touching any production interface.

4.2 VLAN and Subnet Build Workflow

6. Create the VLAN in the switch database using the approved VLAN ID and standard name.
7. Update the VLAN register and subnet inventory before enabling endpoint use.
8. If routing is required, coordinate the Layer 3 gateway, firewall interface, or router subinterface build before cutover.
9. Ensure any DHCP scope, reservation, or helper configuration is ready before production endpoint migration.
10. Label the VLAN purpose and associated subnet in the topology and port map.

VLAN ID	Name	Purpose	Subnet	Gateway
10	SOC-MGMT	Administrative management traffic	192.168.10.0/24	192.168.10.1
20	SOC-LAN	Analyst / workstation network	192.168.20.0/24	192.168.20.1
30	SOC-SIEM	Security monitoring and log systems	192.168.30.0/24	192.168.30.1
40	SOC-SERVER	Internal application / infrastructure servers	192.168.40.0/24	192.168.40.1

The example above is illustrative. The implemented VLAN plan must match the current SOC.lab architecture and approved address plan.

5. Switch Port / Trunk / Access Port Configuration Procedure

5.1 Port Role Definitions

Port Type	Use Case	Standard Rule
Access Port	Single endpoint or host interface	Carry one untagged VLAN only
Trunk Port	Uplink between switches, hypervisor uplink, firewall/router uplink	Carry only explicitly allowed VLANs
Disabled / Unused Port	No active endpoint	Administratively shut down and labeled unused

5.2 Access Port Standard

- Use access mode for end-user devices, servers with a single untagged network, printers, and similar edge systems.
- Assign exactly one approved access VLAN.
- Apply an interface description naming the connected system or room/seat when possible.
- Disable and document any unused ports.

5.3 Trunk Port Standard

- Use trunk mode only when multiple VLANs must traverse the interface.
- Restrict allowed VLANs to only those required; never leave trunks broadly open by default unless explicitly designed.
- Apply descriptions identifying the connected device and link purpose.
- Validate native VLAN behavior according to the switch standard and avoid undocumented native VLAN use.

5.4 Configuration Workflow

11. Identify target switch and interface.
12. Confirm whether the port is an access port or trunk port.
13. Apply the required VLAN assignment or trunk allowance.
14. Save the running configuration according to platform standard.
15. Validate link state, VLAN membership, and endpoint reachability.
16. Update the port map and change record.

Where supported by the platform, port-security, storm control, BPDU guard, and management ACLs should be applied according to the switch baseline standard. These controls are complementary to this SOP and must not be omitted when already required by the environment baseline.

6. Validation and Testing

6.1 Required Validation Checks

Check	Method	Expected Result	Record In

VLAN exists	show vlan / equivalent	Correct VLAN ID and name present	Change record
Access port assignment	show interface switchport / equivalent	Correct access VLAN applied	Port map
Trunk state	show interfaces trunk / equivalent	Trunk up with approved VLAN list only	Change record
Endpoint connectivity	Ping / application test / host check	Endpoint reaches intended gateway/services only	Validation notes
Segmentation control	Cross-VLAN negative test where applicable	Unauthorized access blocked	Security validation notes

7. Troubleshooting

7.1 Common Faults

- Endpoint on wrong VLAN due to incorrect access-port assignment.
- Trunk up but missing one or more required allowed VLANs.
- Subnet built but default gateway or DHCP scope not ready.
- Native VLAN mismatch or undocumented trunk behavior.
- Port description and documentation not updated, causing future operational confusion.

7.2 Troubleshooting Method

17. Verify physical link state first.
18. Confirm VLAN existence and correct port mode.
19. Validate endpoint addressing, gateway, and DHCP behavior.
20. Validate trunk allowed VLAN list and uplink path.
21. Validate any required firewall or router gateway dependency.
22. Compare running configuration against the approved change plan.

8. Change Control and Rollback

8.1 Rollback Standard

Every VLAN or switch-port change must include a rollback path. At minimum, rollback steps must state the prior port mode, prior VLAN assignment, prior trunk allowance list, and any upstream dependency to be restored. If validation fails, the port must be returned to the last known good state and the failure documented.

8.2 Minimum Change Record Content

- Change reason and affected business/security function
- Switch and interfaces affected
- VLAN ID(s), subnet(s), and gateway impact
- Implementation time window and approver
- Validation evidence and rollback result if used

9. Documentation Requirements

- VLAN register
- Subnet inventory
- Switch port map
- Topology diagram when a new segment or uplink changes architecture

- Change control record and validation notes

10. Security Considerations

10.1 Core Security Rules

- Segment networks by trust level and function.
- Keep management traffic separate from user and lab traffic.
- Allow only approved VLANs on trunks.
- Document all exceptions and temporary workarounds.
- Use firewall, ACL, or router controls where cross-VLAN communication must be restricted.

11. Appendix - Standard Command Set

Task	Example Command / Syntax
Create VLAN	`vlan 20` ` name SOC-LAN`

Configure access port	`interface fastEthernet0/3`
	` description Analyst-WS-03`
	` switchport mode access`
	` switchport access vlan 20`
	` spanning-tree portfast`

Configure trunk port	`interface gigabitEthernet0/1`
	` description Uplink-to-Core`
	` switchport mode trunk`
	` switchport trunk allowed vlan 10,20,30,40`

Validate VLANs	`show vlan brief`
Validate trunking	`show interfaces trunk`
Validate interface switchport state	`show interfaces switchport`

Revision History

Version	Date	Author / Owner	Notes
1.0	2026-03-21	Joshua Sartain	Initial combined SOP for VLAN/subnet implementation and switch port configuration.